

Witnessing workplace bullying — A systematic review and meta-analysis of individual health and well-being outcomes

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ABSTRACT

Through a systematic review and meta-analysis of research on individual health and well-being outcomes this paper examines the consequences of witnessing, and thereby being a bystander to, workplace bullying. The review was limited to peer-reviewed primary observational studies with cross-sectional or prospective research design which included findings on outcomes among witnesses to bullying. The review identified 24 relevant studies from 13 countries. Eighty-eight percent of the studies were published from 2010 and onwards. Most studies used cross-sectional single source data from non-probability samples, mainly comprising female respondents from Western countries. Although cross-sectional findings indicated significant associations between witnessing bullying and outcomes such as mental health, job dissatisfaction, and turnover intent, the review show that we need to consider reverse causation, the witness's own exposure to bullying, their proximity and identification with the target, as well as their helping behavior, to understand the true magnitude of the association. Witnessing and being a bystander to bullying is a complex phenomenon and the magnitude of the outcomes relies on a range of third variables and indirect relationships. There is a need for more research to fully understand the consequences of witnessing bullying in the workplace.

1. Introduction

Workplace bullying is defined as a systematic and prolonged mistreatment of one or more employees by supervisors and/or colleagues at the workplace, and where the targets also find it difficult to defend themselves against the harassment (Einarsen, Hoel, et al., 2020b). Bullying behaviors include, but are not limited to, *person-related mistreatment* such as intimidating, undermining, or hostile comments or gestures, social exclusion, spreading of rumors, as well as *work-related mistreatment* such as sabotage of the conditions of work, receiving an unreasonable workload, illegitimately being deprived of responsibilities or work tasks, and unfair criticism of performance (Einarsen & Raknes, 1997; Zapf & Einarsen, 2005). The above definition suggests that bullying represents a process with two distinct steps reflecting perceived “exposure to bullying behavior” and “victimization” (Nielsen et al., 2022).

In the first step, “exposure to bullying behavior”, an employee experiences systematic acts of aggression, social exclusion, and

mistreatment from others at the workplace over an extended time period. In the “victimization” step, the exposed employee feels unable to defend against this exposure due to an experience of imbalance in power. Hence, compared to other forms of mistreatment, such as incivility, abusive supervision, and social undermining, which all may represent single incidences, workplace bullying refers to a form of systematic and prolonged exposure that gradually pushes the target into a helpless situation with less and less ability to avert or stop the negative behaviors (Einarsen, Hoel, et al., 2020a). The concept of workplace bullying should therefore be seen as a continuum, more than an either-or phenomena and this means that there is a significant overlap between workplace bullying and other forms of mistreatment such as harassment, incivility, undermining since these related phenomenon are likely to represent various forms of “bullying behaviors” (Nielsen, Notelaers, & Einarsen, 2021). The present study will therefore investigate the full range of reports of workplace bullying from the occasional exposure to uncivil behaviors at work and up to the more severe long-term victimization processes.

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Although prevalence rates vary extensively between samples and different measurement methods (Leon-Perez et al., 2021; Zapf et al., 2020), bullying is considered a key occupational hazard in contemporary working life (Einarsen, Hoel, et al., 2020b; Keashly et al., 2020). Synthesizing the existing evidence, Nielsen et al. (2010) estimated the global prevalence rate of bullying behaviors to be 15 %, while the prevalence of victimization from bullying was 11 %. In addition to being a prevalent problem, it has been firmly documented through systematic literature reviews and meta-analyses that being a target of this kind of systematic mistreatment at the workplace is associated with reduced affective and attitudinal well-being (Nielsen & Einarsen, 2012), mental and somatic health problems (Nielsen et al., 2014; Verkuil et al., 2015; Xu et al., 2018), sleep problems (Nielsen et al., 2020), symptoms of posttraumatic stress disorder (Nielsen et al., 2015), sickness absence (Nielsen et al., 2016), and suicidal ideation/suicide (Conway et al., 2022; Leach et al., 2017). However, a significant shortcoming of research within the field is that most studies have been limited to studying the outcomes and costs for those individuals directly exposed, whereas less is known about how bullying impacts others at the workplace (Nielsen & Einarsen, 2018; Pouwelse et al., 2018).

As workplace bullying, by definition, occurs within or in the context of a work group or department, there are often more potential bystanders in the actual bullying situations than there are bullies and targets (Pouwelse et al., 2018). As a consequence, bullying may not only be a problem for those directly exposed, but may also have the potential to have indirect ripple effects on others in the work group either through directly witnessing others being perpetrators or targets of bullying, or through second-hand information about episodes of bullying (Salin & Notelaers, 2020). Specifically, building on a “victim by proxy” hypothesis, it has been suggested that observing the bullying of others can be perceived as threatening for the witnesses as it may indicate how any employee at the workplace can be treated (Sprigg et al., 2019). That is, if an employee is treated badly, a bystander may also risk similar negative treatment in the future (Niven et al., 2020). This is in line with social information processing theory which suggests that individuals gather information from their social environments to understand events, form attitudes, and develop and determine expectations on one’s behavior and also its implications (Salancik & Pfeffer, 1978). Similarly, witnessing bullying may provide indications of a poor and detrimental psychosocial working environment which may again lead to reduced job satisfaction and commitment, and increased intent to leave the job (Ozer & Escartin, 2023). Witnessing bullying may also be detrimental to psychological health. For instance, observing the bullying of others without being able to provide support and help may lead to an experience of cognitive dissonance. That is, strong, coexistent and contrasting emotions, which may be distressing and thereby detrimental for a person’s self esteem, health, and wellbeing in the short and long term (Blakey et al., 2023). Specifically, as most humans consider themselves as benevolent and responsible persons (Janoff-Bulman, 1992), the knowledge that one did not try to or felt unable to help a bullied, harassed and mistreated colleague may lead to feelings of shame and guilt due to an injury to the individual’s moral conscience and values. Experiencing dissonance between self-perceptions and actual behavior over a prolonged time may subsequently develop into psychological distress in the form of anxiety and depression (Mikkelsen, 2001). Building on this line of reasoning, it seems likely that helping behavior is a potential moderator of mental health outcomes following witnessed bullying (Nielsen, Rosander, et al., 2021).

Despite having theoretical reasons for expecting bullying to also impact bystanders, narrative reviews of empirical research on the potential ripple effects show that findings appear fragmented and inconclusive (Niven et al., 2020; Pouwelse et al., 2018). That is, while some studies suggest that witnessing bullying is detrimental to health and well-being of the observer (Emdad et al., 2013), other findings question such a relationship (Nielsen & Einarsen, 2013). This inconsistency indicates that there is a need for a systematic synthesis of the evidence that

can elucidate the potential outcomes for bystanders. A systematic review is also needed to shed light on methodological challenges that may have hampered our understanding of how bullying impact witnesses. That is, it has been argued that the effect sizes reported in primary studies are exaggerated due to the fact that many studies on witness reactions have not taken into account (a) the observers own exposure to bullying, and/or (b) adjusted for the observers previous health status (Nielsen & Einarsen, 2013; Sprigg et al., 2019). As for the witnesses’ own exposure to bullying, research has shown that many witnesses have been exposed to bullying themselves (Hauge et al., 2007; Hoel et al., 2010). Consequently, witness reports may be significantly colored by the observer’s own experiences as a victim, and any reported outcome could therefore be a consequence of exposure to, rather than observation of, bullying (Nielsen & Einarsen, 2013). Regarding the role of the observer’s previous health status, it has been suggested that existing health problems may influence observations of bullying. Following the “gloomy perception mechanism” (de Lange et al., 2005), mentally distressed employees may report less favorable work characteristics because they evaluate their work environment more negatively than do other employees and therefore have a lower threshold for attributing and labeling negative events at the workplace as bullying. Hence, an association between witnessing bullying and outcomes may not be straightforward and clear-cut but is likely to be influenced by a range of different factors.

Taken together, the actual magnitude of the outcomes of observed workplace bullying on health and well-being among witnesses is still unknown. To fill this knowledge gap, and thereby to move the field forward, the overarching aim of this systematic review and meta-analytic study is to provide a detailed and comprehensive synthesis of available primary studies on individual health and well-being outcomes among witnesses and bystanders of bullying at the workplace. To guide our understanding of how bullying may influence witnesses, the review will determine:

- 1) The health and well-being outcomes that have been examined in research on witnesses to bullying so far, and thereby also which outcomes that have not been investigated.
- 2) The methodological designs that have been used in research on witnesses of bullying.
- 3) The magnitude of the associations between witnessing bullying and the health and well-being outcomes.
- 4) The theoretical frameworks that have been used to explain the impact of witnessing bullying on outcomes.
- 5) The mediators and moderators that have been accounted for in research.
- 6) Whether the health and well-being of employees predicts subsequent risk of witnessing bullying of others.

This systematic review and meta-analysis contribute to research on aggression and violent behavior by being the first comprehensive and exhaustive statistical synthesis and summary of the empirical evidence regarding the outcomes of witnessing workplace bullying. A systematic synthesis of the literature will provide researchers with an overview of what is already done and known in the field, while also pointing to the knowledge gaps. The current review will therefore provide important guidelines for the direction of future research and will also be highly relevant for the practice field.

2. Methods

This review has been written in accordance with the PRISMA-P (Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols) guidelines and the MOOSE Guidelines for Meta-Analyses and Systematic Reviews of Observational Studies (Moher et al., 2015; Stroup et al., 2000). The review was preregistered in Prospero (No: CRD42022342006) and a protocol describing the background and research approach has been pre-published (Nielsen et al., 2023).

2.1. Literature search

Relevant articles were identified through systematic searches in the following databases: Medline/Pubmed, Proquest, Web of Science, Taylor & Francis Online Journals, PsycInfo, and Wiley Online Library. Additional searches were performed in Google Scholar. The literature search was not limited by historical time-constraints. The search terms were adopted from previous systematic reviews on workplace bullying (Nielsen et al., 2015; Nielsen et al., 2016; Nielsen et al., 2020) and narrative reviews on bystanders to bullying (Niven et al., 2020; Pouwelse et al., 2018) and adjusted to fit the topic of the current review. An initial scoping search with no restrictions or limitations was conducted using a combination of free-text search terms. Returned search results were reviewed to identify potentially relevant subject headings, free-text terms, and phrases. The final search strategies were constructed from combinations of Medical Subject Heading (MeSH) terms and keywords/free-text terms, adjusted for each database. The final utilized search terms are presented in Table 1. The systematic searches were conducted by combining every possible combination of the three categories of keywords. No specific search terms were included for the outcome variables as the aim of the review was to identify every health and well-being outcome examined among witnesses of bullying to this date. Information about specific outcomes of witnessing bullying were therefore extracted manually during the screening of included studies.

The literature search was conducted by a professional librarian with extensive experience regarding systematic reviews in September 2022. The search results were then exported to Covidence, a web-based screening and data extraction tool for authors conducting systematic reviews. After removing duplicates, studies were independently screened by the first and the last author based on title and abstract to establish whether it was likely to provide information on outcomes among witnesses to workplace bullying. The full texts of remaining studies were then manually screened, again independently, based on the inclusion and exclusion criteria described below. In case of inconsistencies, classification mismatches were discussed and resolved by consensus. The reference lists of included full text articles were manually inspected to detect studies that were missing in the literature search.

2.2. Eligibility criteria

Eligible studies should report empirical findings on any individual health and well-being outcome variable assessed among witnesses (observers or bystanders) to workplace bullying or any overlapping concept such as harassment, incivility, social undermining, abusive supervision etc. Although there may be some definitional differences and nuances between bullying and these other concepts, research on targets show that the consequences are overlapping (Hershcovis, 2011; Raver & Barling, 2007). In addition, as explained in the introduction of this study, the types of workplace mistreatment included in our search terms

can all represent typical forms of bullying behaviors and can therefore be perceived by witnesses as bullying. “Well-being” refers to the various life/non-work satisfactions enjoyed by individuals, work/job-related satisfactions, and general health, whereas “health” reflects a sub-component of well-being that comprises the combination of mental/psychological indicators and physical/physiological indicators (Danna & Griffin, 1999), including psychosomatic indicators.

The review was limited to primary observational studies with cross-sectional or prospective research designs, and case-control studies which included findings on witnesses to actual bullying. Qualitative interview studies, experimental studies, single case studies, and series of single case studies were excluded. We also excluded primary observational studies in which it was not possible to extract relevant data on the association between being a witness to bullying and outcomes. The study population was restricted to adults (18 years or older) with a current or previous employment in a full- or part-time position. We considered workers in any trade, industry, and occupation. No restrictions were placed regarding gender, ethnicity, or other demographic characteristics of the subjects. Articles published in peer-reviewed journals written in English, German, French, Spanish, or the Scandinavian languages (Danish, Norwegian, and Swedish) were included in the literature review, whereas articles in other languages, conference abstract, dissertations, gray literature, and unpublished literature were not included. Hence, the present review concerns published peer reviewed studies only.

To be included in the meta-analytic part of the review, studies were required to provide the zero-order associations between witnessing bullying and outcomes or provide sufficient information for these associations (effect sizes) to be calculated. Studies lacking this information or reported effect sizes that could not be transformed into correlations were excluded from the meta-analysis. To avoid double-counting data, each sample was only used once when it was used in more than one study included in the review. In cases with overlap, we used data from the largest sample.

2.3. Data extraction

Using a standardized form, the first and the last author independently excerpted data on reference information (authors, publication year, journal, title); sample characteristics; geographical origin of study, study design, theoretical framework for the study, number of participants, response rate, gender distribution, mean age, sampling procedure, tools for measuring workplace bullying and witnessing workplace bullying, health and well-being outcomes, effect sizes for association between witnessing bullying and outcomes, whether the witness own exposure to bullying was adjusted for, moderators and mediators examined, and overlap in sample with other studies in the review.

2.4. Assessment of methodological quality (risk of bias)

The methodological quality of the included studies was assessed with an adapted version of a previously established checklist for research on workplace bullying comprising 11 items related to sampling, representativeness, measurement issues, and confounders (Nielsen et al., 2016; Nielsen et al., 2020). This checklist comprises selected and adapted items from the Risk of Bias Assessment Tool for Nonrandomized Studies (Kim et al., 2013) and the Quality Assessment Tool (National Collaborating Centre for Methods and Tools, 2008). The quality of the reviewed studies will be scored on a scale from 0 (lowest possible quality) to 11 (highest possible quality).

2.5. Meta-analytic approach

The Comprehensive Meta-Analysis (version 2) software developed by Biostat (Borenstein et al., 2005) was used to conduct the meta-analysis. Correlations with 95 % Confidence Intervals (95 % CI) were

Table 1
Search terms used to identify relevant studies.

Group 1	Group 2	Group 3
Work*	Bullying	Witness*
Job	Mobbing	Observ*
Occupation*	Victimization	Bystand*
Employee	Emotional abuse	Third-part*
Organization*	Incivility	Whistleblow*
	Psychological aggression	Co-worker
	Mistreatment	Colleague
	Ostracism	
	Exclusion	
	Undermining	
	Harassment	
	Cyberbullying	
	Cyberharassment	

* Indicates MESH-terms.

calculated to determine the overall magnitude of the synthesized effect size. Hedges' g were reported to indicate the size of the effect: Small effect = 0.2, medium effect = 0.5, and large effect = 0.8. In studies where several effect sizes were reported from the same sample, the mean of the combined effect sizes was calculated (e.g., models with different control variables). In studies reporting effect sizes from independent subgroups (e.g., moderators), each subgroup was included as a unique sample in the meta-analysis. Moderation analyses were also used to compare the magnitude of the associations in cross-sectional and

prospective data.

The Comprehensive Meta-analysis program weights studies by inverse variance. Inverse-variance weighting is a method of aggregating two or more random variables where each random variable is weighted in inverse proportion to its variance to minimize the variance of the weighted average. The inverse variance is roughly proportional to sample size, but is a more nuanced measure, and serves to minimize the variance of the combined effect (Borenstein et al., 2007). The pooled mean effect size was calculated using a random effects model. Random

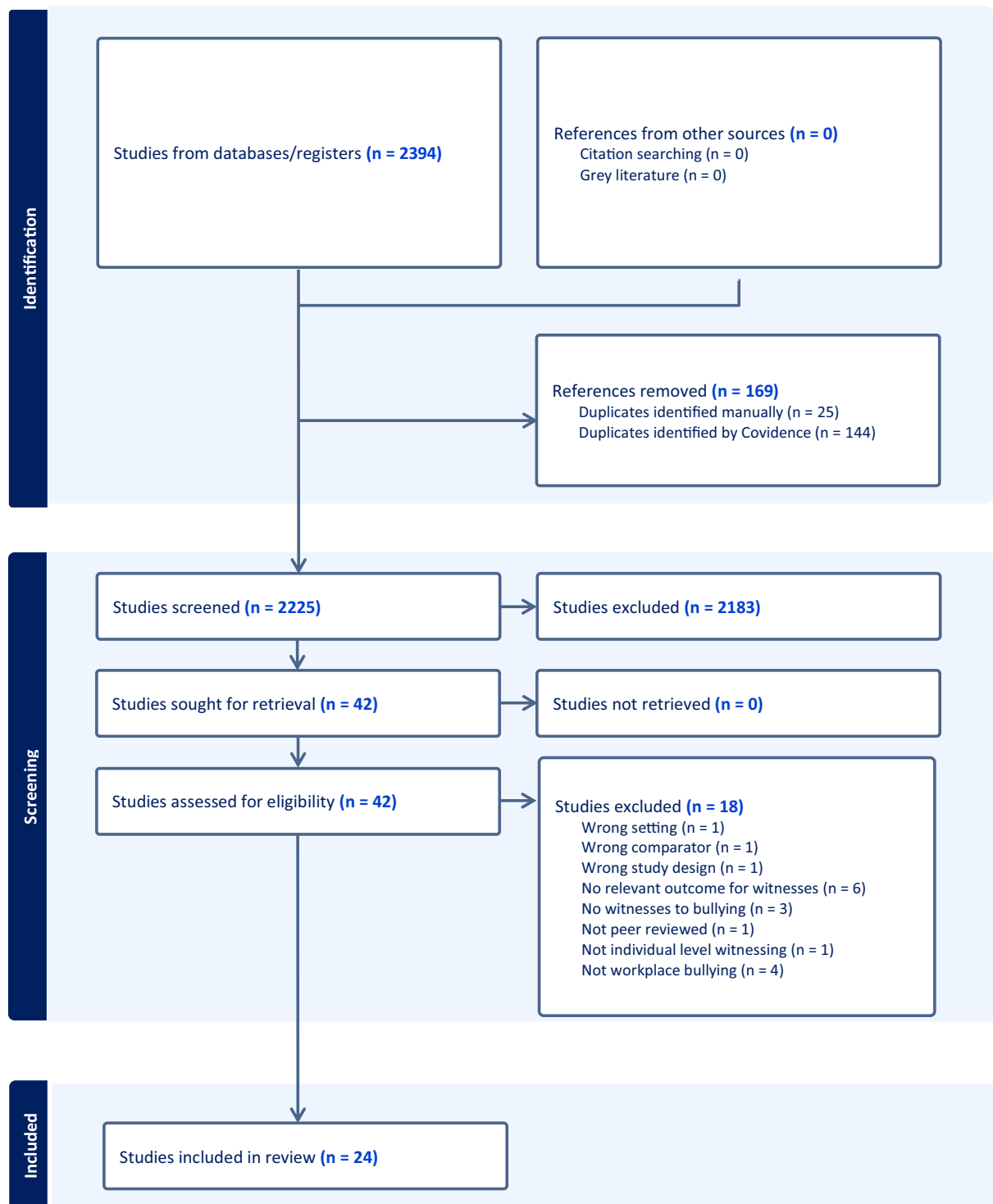


Fig. 1. Flow diagram of study selection. Adapted from PRISMA statement guidelines.

effects models are recommended when accumulating data from a series of studies where the effect size is assumed to vary from one study to the next, and where it is unlikely that studies are functionally equivalent (Borenstein et al., 2007). Random effects models allow statistical inferences to be made to a population of studies beyond those included in the meta-analysis (Berkeljon & Baldwin, 2009). The Q_{within} -statistic was used to assess the heterogeneity of studies, and the I^2 -statistic was computed to determine heterogeneity in terms of percentages. Increasing values show increasing heterogeneity, with values of 0 % indicating no heterogeneity, 50 % indicating moderate heterogeneity, and 75 % indicating high heterogeneity, respectively (Higgins et al., 2003). We examined three different indicators of publication bias: Funnel Plot, Duval and Tweedie's trim and fill procedure, and Egger's Regression Intercept (Borenstein et al., 2009).

3. Results

In total, 2394 studies were identified in the electronic literature search. Removing duplicates, 2225 non-overlapping manuscripts were retained for further screening. After screening of titles and abstracts, 42 manuscripts were included in the full-text assessment. Of these, 18 were excluded due to reasons such as having a non-relevant or missing outcome variable, missing data in witnesses to bullying, or an irrelevant study setting. Consequently, 24 studies met the inclusion criteria and were included in the qualitative synthesis. A flow diagram of the study selection process, adapted from PRISMA statement guidelines, is depicted in Fig. 1.

3.1. Characteristics of the included studies

The 24 included studies, one with two samples (Salin & Notelaers, 2020), and their characteristics, are presented in Table 2. Altogether 16 of the studies examined "workplace bullying" as the key constructs, while four studies focused on "incivility". The remaining studies examined variations of "harassment", "discrimination" or "unpleasant interactions". The samples originated from 13 different countries (Belgium, China, Cyprus, Denmark, Finland, Italy, Lithuanian, New Zealand, Norway, Spain, Sweden, UK, USA) and 88 % of the studies were published in the period 2010 to 2023 (Range: 2001–2023). The average number of respondents across the 25 samples was 1073 persons (Range: 37–5388). Altogether 19 studies reported response rate, with a mean rate of 46.4 %. Regarding gender distribution, females comprised the majority in 22 of 25 samples with an average of 65.13 % females in the samples. The mean age in the samples was 41.7 years. These descriptive findings indicate that most research on bystanders to bullying has been carried out in Western Europe and USA during the last decade and that most participants in studies on bystanders are female.

3.2. Methodological designs

Eighteen of the samples comprised cross-sectional data, six used prospective data with two time-points, and one sample was from a quantitative diary study. Based on the criteria for random and non-random sampling procedures suggested by Ilies et al. (2003), 14 of 25 samples were based on non-probability sampling procedures, while 11 studies used some sort of randomization procedure to sample respondents. Seven studies used a scale-based inventory (e.g., multiple items) to measure witnessed bullying, ten studies used a single item question that presented the respondents with a definition of the workplace bullying construct before asking whether they had witnessed this kind of mistreatment. Eight studies used a single item question without any preceding definition of the workplace bullying construct.

3.3. Quality assessment

On a scale from 0 to 11, the methodological quality of the reviewed

studies ranged from 1 to 9 with a mean score of 4.52 and a median of 4. The quality assessment indicated that most studies had multiple inadequacies regarding the methodological approaches, and that the risk of bias related to selection of respondents, representativeness, measurement, and confounders were moderate to high.

3.4. Outcomes examined

Table 3 provides an overview of the different outcomes that have been examined in studies of witnesses to bullying to this date. As for indicators of health, mental health complaints (e.g., anxiety and depression) were examined in 13 studies, and was thereby the most frequently examined outcome of witnessing bullying. Emotional exhaustion/burnout and sleep were examined in four studies each, whereas somatic complaints were included in three studies. Other indicators of mental and somatic health that have been examined were symptoms of posttraumatic stress disorder, overall emotional well-being, and use of sleep-inducing drugs/sedatives (1 study each). Examined in eight studies, job satisfaction was the most frequently examined indicator of behavioral/attitudinal outcomes of witnessing bullying, followed by intent to leave (6 studies), commitment (4 studies), and work disengagement (1 study). Other examined outcomes were strain/stress (5 studies), cortisol (1 study), recovery need (1 study), and worrying (1 study).

3.5. Theoretical frameworks

Eleven of the 24 studies included in the review provided an explicit *a priori* theoretical framework for explaining the association between being a witness to workplace bullying and health outcomes before presenting their results. Four out of six studies that used a theoretical framework for explaining the impact of witnessing on health-related outcomes (mental health, somatic health, exhaustion/burnout, and sleep) were based on some variation of Lazarus and Folkman's (1984) transactional model of stress and coping (Hansen et al., 2006; Miner & Eischeid, 2012; Nielsen, Rosander, et al., 2021; Sprigg et al., 2019). One study used Twemlow et al.'s (2005) model for triadic interaction (Emdad et al., 2013), while one study used a combination of the affective events theory and emotion regulation theory (Totterdell et al., 2012). Although it has been argued that victim by proxy, also referred to as secondary trauma theory, has been suggested as a key explanation for the health outcomes among bystanders (Niven et al., 2020), this theory was only highlighted in three studies (Nielsen, Rosander, et al., 2021; Rosander & Nielsen, 2023; Sprigg et al., 2019).

No consistent model or theory was used to explain the potential impact of witnessing bullying on well-being outcomes: Miner and Cortina (2016) used a moderated mediation model developed for the study to explain associations with job satisfaction and intent to leave, while Morrow et al. (2011) built their reasoning for an association between witnessing bullying and job satisfaction on social exchange theories and norms of reciprocity. Salin and Notelaers (2020) referred to psychological contract violation as a mechanism that would explain associations between witnessing bullying and the following outcomes: job satisfaction, commitment, intent to leave, sleep, worrying, and recovery need, while Sims and Sun (2012) used organizational strain theory in combination with withdrawal theory to determine associations with job satisfaction, commitment, intent to leave, and strain/stress. As the present review only aimed to provide an overview of the utilized theoretical frameworks, the reader should refer to the individual studies for a further explanation of the theoretical models used in research.

3.6. Moderators and mediators

Table 4 presents an overview of moderators and mediators examined in research on outcomes of witnessing bullying at the workplace. A moderator is a variable that affects the strength and direction of a

Table 2
Study characteristics.

Reference	Country	Design	Concept examined	N	Response rate	Sampling method	Assessment method bystanding	Outcomes	Adjusted for exposure in analyses	Quality 0–11
Aristidou et al. (2020)	Cyprus	Cross-sectional	Bullying	113	32.56	Non-random	SL w/def	- PTSD - Burnout/exhaustion - Job satisfaction - Sleep	Yes	2
Arnetz et al. (2019)	USA	Cross-sectional	Bullying	331	24.3	Random	SL w/def	- Work engagement	Yes	5
Astrauskaite et al. (2010)	Lithuania	Cross-sectional	Colleague harassment	351	NA	Non-random	SL wo/def	- Burnout/exhaustion	No	1
Bentley et al. (2012)	New Zealand	Cross-sectional	Bullying	332	15	Random	SL wo/def	- Mental health - Commitment - Intent to leave - Overall emotional well-being	No	3
Cardoso et al. (2016)	Spain	Cross-sectional	Psychological harassment	204	NA	Random	SL wo/def	- Mental health - Intent to leave - Strain/stress - Sleep	Yes	3
Emdad et al. (2013)	Sweden	Longitudinal	Bullying	2177	66.23	Non-random	SL w/def	- Mental health	No	6
Hansen et al. (2014)	Denmark	Longitudinal	Bullying	3382	46	Non-random	SL w/def	- Sleep	No	4
Hansen et al. (2006)	Sweden	Cross-sectional	Bullying	437	70	Non-random	SL w/def	- Mental health - Somatic health - Cortisol	Yes	5
Hoel et al. (2004)	UK	Cross-sectional	Bullying	5388	42.8	Non-random	SL w/def	- Mental health - Somatic health	Yes?	3
Holm et al. (2019)	Sweden	Cross-sectional	Incivility	978	21.23	Random	Behavioral	- Job satisfaction - Strain/stress	No	3
Maran et al. (2021)	Italy	Cross-sectional	Mobbing	574	60.7	Non-random	SL w/def	- Mental health - Strain/stress	No	2
Miner and Cortina (2016)	USA	Cross-sectional	Incivility	1702	67	Random	Behavioral	- Job satisfaction - Intent to leave	No	6
Miner and Eischeid (2012)	USA	Cross-sectional	Incivility	453	NA	Non-random	Behavioral	- Mental health	No	3
Morrow et al. (2011)	USA	Cross-sectional	Incivility	1381	45.1	Random	Behavioral	- Job satisfaction	Yes	6
Nielsen and Einarsen (2013)	Norway	Longitudinal	Bullying	1775	57	Random	SL w/def	- Mental health	Yes	8
Nielsen, Rosander, et al. (2021)	Sweden	Longitudinal	Bullying	1096	25	Random	SL wo/def	- Mental health	Yes	8
Rhead et al. (2020)	UK	Cross-sectional	Discrimination and harassment	931	NA	Non-random	SL wo/def	- Mental health - Somatic health	No	3
Rosander and Nielsen (2023)	Sweden	Longitudinal	Bullying	788	25	Random	SL wo/def	- Mental health	Yes	9
Salin and Notelaers (2020) Sample 1	Belgium	Cross-sectional	Bullying	1473	70	Non-random	SL w/def	- Job satisfaction - Commitment - Intent to leave - Sleep - Worrying - Recovery need	Yes	6
Salin and Notelaers (2020) Sample 2	Finland	Cross-sectional	Bullying	1148	26.2	Non-random	SL w/def	- Job satisfaction - Commitment - Intent to leave	Yes	6
Sims and Sun (2012)	China	Cross-sectional	Bullying	150	75	Non-random	Behavioral	- Job satisfaction - Commitment - Intent to leave - Strain/stress	No	2
Sprigg et al. (2019)	UK	Longitudinal	Bullying	194	NA	Non-random	Behavioral	- Mental health - Burnout/exhaustion	Yes	4

(continued on next page)

Table 2 (continued)

Reference	Country	Design	Concept examined	N	Response rate	Sampling method	Assessment method bystanding	Outcomes	Adjusted for exposure in analyses	Quality 0–11
Totterdell et al. (2012)	UK	Quantitative diary	Unpleasant interactions	37	67.3	Non-random	SL wo/def	- Burnout/exhaustion	No	4
Vartia (2001)	Finland	Cross-sectional	Bullying	949	65.5	Random	SL wo/def	- Mental health - Strain/stress - Sleep inducing drugs	Yes	6
Wadsworth and Estrada (2022)	USA	Cross-sectional	Harassment	482	21	Random	Behavioral	- Job satisfaction	No	5

Table 3

Outcomes examined in studies on bystanders to bullying.

Outcome	No. of studies
Mental health	13
Job satisfaction	8
Intent to leave	6
Strain/stress	5
Burnout/exhaustion	4
Commitment	4
Sleep	4
Somatic health	3
Cortisol	1
Overall emotional well-being	1
Posttraumatic stress symptoms	1
Recovery need	1
Sleep inducing drugs	1
Work disengagement	1
Worrying	1

relationship between an independent and a dependent variable (Baron & Kenny, 1986). Being included in four different studies, gender was the most frequently examined moderating variable. In a study from Italy, Maran et al. (2021) reported that female witnesses reported more severe mental health complaints compared to male witnesses. Similar results were established in a longitudinal study with data from four large industrial enterprises in Sweden with the findings showing that although the number of men who were bystanders to bullying was larger compared to women, the proportion of women who were bystanders to bullying and developed depressive symptoms 18 months later was higher in comparison with men (Emdad et al., 2013). Being a woman was also a risk factor in a study of observed workplace incivility toward women among 1702 US employees (Miner & Cortina, 2016). The findings showed that although both male and female bystanders reported heightened turnover intentions with higher levels of observed incivility toward women, these relationships were stronger for female than male observers. In addition, witnessing incivility toward female coworkers

was only related to lower job satisfaction among female bystanders. Finally, in a study of negative emotions as outcomes of observed incivility, participants reported more negative emotionality when they witnessed incivility toward same-gender coworkers compared to opposite-gender coworkers. This relationship was especially pronounced for male observers for anger, fear, and anxiety, and female observers for demoralization (Miner & Eischeid, 2012).

Active and passive roles when being a bystander may also contribute to the magnitude of outcomes. A longitudinal probability survey of 1096 employees from the Swedish workforce, with an 18-month time lag between assessment points, showed that intervening against bullying moderated the relationship between observations of bullying and mental health problems (Nielsen, Rosander, et al., 2021). Observers who did not try to intervene against the observed bullying reported a significant increase in mental health problems at follow-up, whereas there were no significant changes in levels of mental health problems among those who did try to intervene. Adding to the understanding of the bystander–target relation, a UK quantitative diary study in which participants recorded their reactions to over 1000 interpersonal interactions between coworkers across 15 work days showed that staff felt more emotionally drained when they took the perspective of the target to a greater extent and when they witnessed interactions directly (first-hand) rather than indirectly (Totterdell et al., 2012).

Access to personal and contextual resources may also be significant in our understanding of outcomes. Sprigg et al. (2019) found, in a two-wave study involving 194 employees, that witnessing bullying was associated with depression and anxiety 6 months later, but only if the employees were low in optimism and lacked supervisor support. In addition, the experience of strong coworker support weakened the relationship between witnessing bullying and emotional exhaustion and work-related depression as outcomes.

A mediator is a variable that explains the process through which an independent and a dependent variable is related (Baron & Kenny, 1986). As displayed in Table 4, several potential mediators have been examined regarding the association between witnessing bullying and outcomes. In

Table 4

Moderators and mediators examined in research on outcomes of witnessing bullying.

Reference	Study design	Moderators	Mediators	Outcome
Maran et al. (2021)	Cross-sectional	Gender		Mental health
Emdad et al. (2013)	Longitudinal	Gender		Mental health
Hansen et al. (2006)	Cross-sectional		Social support	Mental health
Miner and Cortina (2016)	Cross-sectional	Gender	Perceptions of injustice	Job satisfaction Turnover intentions
Miner and Eischeid (2012)	Cross-sectional	Gender		Mental health
Nielsen, Rosander, et al. (2021)	Longitudinal	Intervening against the bullying		Mental health
Salin and Notelaers (2020)	Cross-sectional		Psychological contract violation	
Sims and Sun (2012)	Cross-sectional		Strain	
			Job satisfaction	
Sprigg et al. (2019)	Longitudinal	Trait optimism Social support from coworkers Supportive supervisory style		Mental health
Totterdell et al. (2012)	Quantitative diary	Take perspective of target Witnessing mistreatment directly vs indirectly	Negative mood Emotion regulation	Exhaustion

a study of 437 Swedish employees, the findings indicated that the relationship between witnessing bullying and anxiety was fully mediated by low levels of social support from supervisors. Based on this finding, the authors suggested that it may not so much be the observed bullying as the lack of support that explains the association between witnessing bullying and mental health complaints. However, following this line of reasoning, it seems like it may be more reasonable to test social support as a moderating variable.

Justice and fairness are important to people in general, in everyday interactions, and in the workplace, and according to fairness theory, perceptions of injustice at the workplace will have implications for well-being (Folger, 1993). Testing injustice as a potential explanatory mechanism for the association between witnessing incivility and outcomes, Miner and Cortina (2016) showed that perceptions of injustice partially mediated the negative effect of observed incivility toward women on observers' job satisfaction and turnover intentions, thus indicating that witnessing mistreatment of others at the workplace violate the observers experience of justice, and that this may subsequently lead to reduced job satisfaction and increased turnover intent. Examining the related construct of psychological contract violation (employees' feelings of disappointment arising from their belief that their organization has broken its psychological contract of work-related promises) as a mediator, Salin and Notelaers (2020) found evidence for this kind of violation as an explanatory mechanism in the association between witnessing bullying and employee attitudes (job satisfaction, organizational commitment, and turnover intent). Examining a multiple mediator model, Sims and Sun (2012) found that strain and job satisfaction partially mediated the association between witnessing bullying and turnover intent, and that the mediator model explained 33 % of the variance in turnover intent. Finally, in Totterdell et al.'s (2012) diary study comprising data on more than 1000 interpersonal interactions between hospital staff, a moderated mediation analysis showed that witnesses felt more emotionally drained after witnessing unpleasant interactions than pleasant ones, and that this effect was mediated by their negative mood and by the extent to which they had to control their feelings. The effect was amplified when witnesses took the perspective of the target and when they witnessed the interaction directly.

3.7. Reverse causality

Three of the reviewed studies with prospective designs examined so-called reverse causality, that is whether the existing levels of the outcome variable at baseline predicted the likelihood of witnessing bullying at follow-up. In a national representative study of 1775 Norwegian employees with a two year time-lag, it was found that baseline levels of mental distress significantly predicted new cases of observers (OR = 1.68; 95 % CI = 1.12–2.54) when controlling for baseline observations of bullying (Nielsen & Einarsen, 2013). This relationship remained significant even after adjusting for the respondents' own exposure to bullying at baseline. Similar findings were obtained in a representative study of 1096 Swedish employees with an 18 month time-lag (Nielsen, Rosander, et al., 2021). Specifically, the findings showed a significant association between baseline mental health problems and subsequent observations of bullying at follow-up (OR = 1.85; 95 % CI 1.34 to 2.55) controlling for age, gender, and roles in organization, as well as adjusting for observations of bullying at baseline. Contrasting the findings from the two studies above, a UK study of 194 respondents with a six-month time-lag found no indications of depression, anxiety, and emotional exhaustion, respectively, as predictors of subsequent risk of witnessing bullying at follow-up. The analyses were adjusted for age, gender, organizational tenure, witnessed bullying, and experienced bullying at baseline.

The magnitude of associations between witnessing bullying and outcomes: Meta-analysis.

Results from the meta-analyses are displayed in Table 5 (Forest plots for all meta-analyses are included in Appendix 1). The estimates are

based on unadjusted bivariate relations between witnessing bullying and examined outcomes. Only outcomes that were examined in at least two primary studies were included in the meta-analyses. The average Pearson correlation from the 15 studies that provided usable data on the association with mental health complaints was 0.18 ($p < 0.001$; 95 % CI 0.12–0.25). A Hedges' g of 0.42 indicates a small to medium effect size for the relation with mental health complaints. The studies had high levels of heterogeneity ($Q_{\text{within}} = 87.51$; $p < 0.001$; $I^2 = 84.00$). A funnel plot indicated relative symmetry among the included studies, thus suggesting that the established estimate is comparable to the population effect. The Duval and Tweedie's trim and fill procedure indicated that there were no missing studies to the right or the left of the mean. Following Egger's regression test, the intercept was not significantly different from zero (Intercept = 0.80; Standard Error = 2.23; 95 % CI –4.00–5.61), thereby indicating that the estimate is not likely to be influenced by potential publication bias. A significant difference emerged when comparing studies that adjusted for the observers own exposure to bullying with studies that did not adjust for own exposure ($Q_{\text{between}} = 7.57$; $df = 1$; $p < 0.01$). The findings showed that the effect size between witnessing bullying and mental health complaints were smaller in studies that adjusted for the observers own exposure ($r = 0.08$; 95 % CI 0.04–0.12; $p < 0.001$) compared to studies without this kind of adjustment ($r = 0.19$; 95 % CI 0.12–0.26; $p < 0.001$).

As for other outcomes, data from three studies showed a small, but significant, association between witnessing bullying and somatic health complaints ($r = 0.07$; 95 % CI 0.01–0.13; $p < 0.05$; Hedges' $g = 0.13$), while a medium association was found for the association between witnessing bullying and sleep problems across two studies ($r = 0.27$; 95 % CI 0.04–0.48; $p < 0.05$; Hedges' $g = 0.58$). Witnessing bullying had a medium to strong negative association with job satisfaction ($r = -0.33$; 95 % CI –0.42 to –0.22; $p < 0.05$; Hedges' $g = -0.73$), a medium negative association with commitment ($r = -0.21$; 95 % CI –0.28 to –0.13; $p < 0.05$; Hedges' $g = -0.48$), and a medium positive association with turnover intent ($r = 0.23$; 95 % CI 0.15–0.31; $p < 0.05$; Hedges' $g = 0.53$). There were no indications of publication bias for somatic health complaints, sleep problems, job satisfaction, whereas the Duval and Tweedie's trim and fill procedure indicated one missing study to the left of the mean for the analyses of turnover intent and commitment. Due to the small number of studies for each outcome, moderator analyses were not possible for somatic health complaints, sleep problems, job satisfaction, turnover intent, and commitment.

4. Discussion

The goal of this systematic review and meta-analysis was to understand how witnessing bullying of others at the workplace influences the health and well-being of the observers. Based on findings from 24 primary studies that met the inclusion criteria, our quantitative synthesis indicates that witnessing bullying is significantly associated with mental and somatic health problems, sleep problems, as well as low well-being, among the witnesses. However, as shown by the moderator analyses and the qualitative synthesis of this body of research, associations between witnessing bullying and outcomes should be interpreted with caution as there are several moderating and confounding variables that are likely to influence the associations. In addition, as shown in the overview of methodological aspects of the research, most studies have low methodological quality and are biased by an overuse of cross-sectional single source data from non-probability samples that mainly comprise female respondents from Western countries. Hence, aspects related to methodology and design must also be considered. Finally, our review of theoretical frameworks shows that there is no coherent theoretical model that can explain how witnessing bullying relates to the examined outcomes, thus indicating that the field of research needs a better theoretical foundation. In the following, we will discuss the limitations of previous research, propose a conceptual model based on the findings from the review (see Fig. 2), and in doing so, suggest a way forward for

Table 5

Meta-analytic findings on the magnitude of the association between witnessing workplace bullying and outcomes (Random effects model).

Outcome	Effect size						Heterogeneity			Publication bias	
	K	N	Hedges' g	Pearson's r	95 % CI r	Q _{within}	I ²	Tau ²	Tau	Egger's test	Trim and fill
Mental health [†]	15	9677	0.42	0.18***	0.12–0.25	87.51	84.00	0.01	0.12	0.80 ^{NS}	0
Somatic health	3	1546	0.13	0.07*	0.01–0.13	0.24	0.00	0.00	0.00	.49 ^{NS}	0
Sleep problems	2	3586	0.58	0.27*	0.04–0.48	164.08	98.78	0.04	0.21	–5.49 ^{NS}	0
Job satisfaction	7	6458	–0.73	–0.33***	–0.42 to –0.22	111.45	94.62	0.02	0.14	1.63 ^{NS}	0
Turnover intent	4	3332	0.53	0.23***	0.15–0.31	12.56	76.12	0.01	0.07	0.53 ^{NS}	1
Commitment	4	1961	–0.48	–0.21***	–0.28 to –0.13	6.73	55.40	0.00	0.06	–2.21 ^{NS}	1

K = Number of effect sizes; N = Number of respondents; OR = Odds Ratio; 95 % CI = 95 % Confidence Interval; r = Pearson correlation.

[†] The mental health category includes studies on mental distress, burnout, and emotional well-being.

upcoming research on outcomes among witnesses to bullying.

4.1. Implications for theory and research

While research to this date has examined a range of different outcomes related to witnessing bullying at the workplace, our review shows that most existing studies have focused on health-related outcomes such as anxiety, depression, and burnout, whereas less is known about attitudinal outcomes such as job satisfaction, turnover intent, and commitment, and psychological outcomes such as sleep, worrying, and rumination. As suggested by theoretical models such as the Transactional model of stress and coping (Lazarus & Folkman, 1984) and the Cognitive activation theory of stress (Ursin & Eriksen, 2004), health outcomes following psychological stressors are assumed to be the result of cognitive activation, such as worrying/rumination, and further that sustained cognitive activation could, over time, lead to impairment of health and well-being. This means that health outcomes, be it mental or somatic, need time to develop and that we need to understand the intermediate mechanisms that can explain how witnessing bullying could influence the health of the witnesses. A continued experience of cognitive dissonance between one's own value system and empathy with the victim, combined with the lack of possibilities to effectively intervene and the stop the bullying, may indeed be such a mechanism for sustained cognitive activation. In line with such an assumption, it has been shown that the relationship between witnessing bullying and health complaints

disappears when controlling for whether the observer has intervened in the situation (Nielsen, Rosander, et al., 2021).

As displayed in our conceptual model (Fig. 2), we propose that both attitudinal and psychological reactions to witnessing bullying are important mediating mechanisms for health and well-being outcomes in this regard. Hence, more research is needed on attitudinal and psychological reactions as direct outcomes of witnessing bullying, as well as on their explanatory role in relation to more distal health outcomes. As for distal outcomes, a noteworthy finding from the review is that we were not able to identify studies that have examined whether witnessing bullying is associated with work related outcomes such as work ability, sickness absence, and actual turnover. Hence, to fully understand the impact that witnessing bullying may have on the observer, and to determine the costs and long-term consequences of witnessing workplace bullying, there is also a need for more research on the above-mentioned work-related outcomes.

Going back to the Transactional model of stress and coping (Lazarus & Folkman, 1984) and the Cognitive activation theory of stress (Ursin & Eriksen, 2004), both models highlight the importance of taking moderating factors into consideration when explaining health outcomes of psychological stressors. For instance, the transactional stress model suggests that the severity of reactions following exposure to a given stressor is a function of a dynamic interplay between event characteristics and individual appraisal. This assumption gains support from the quantitative diary study by Totterdell et al. (2012) which found that

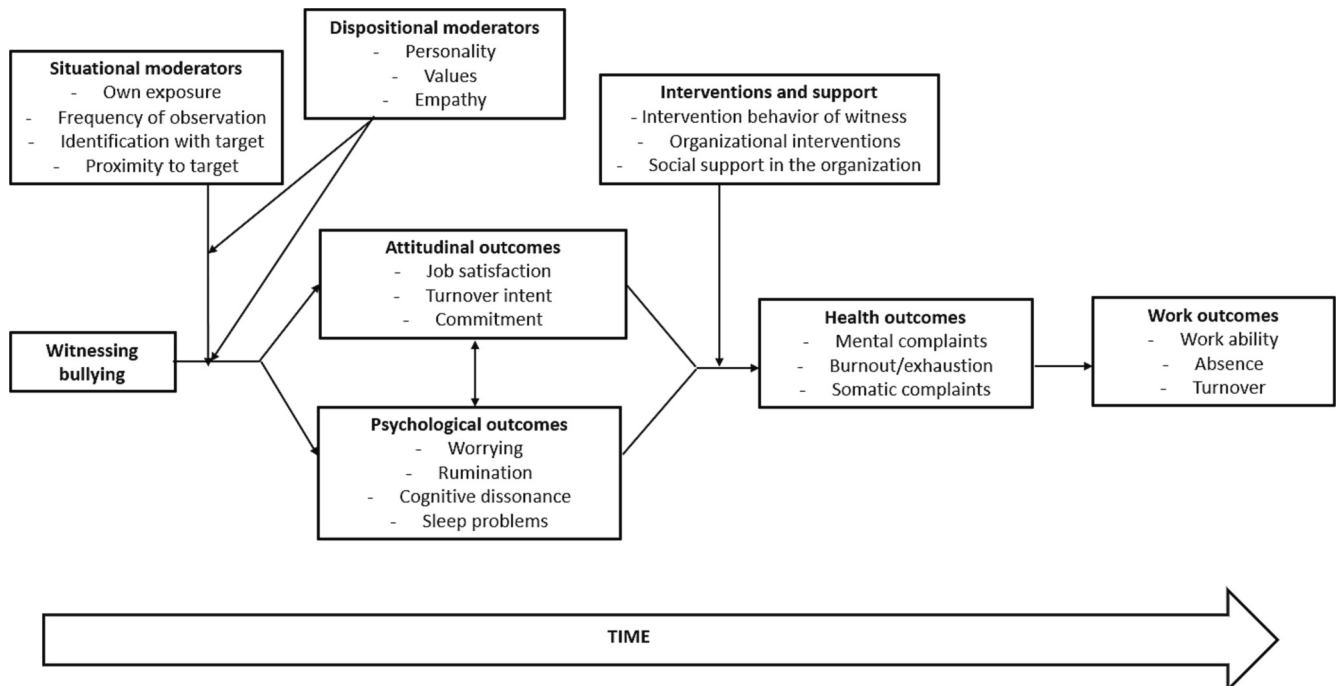


Fig. 2. Conceptual model for research on witnesses to bullying in the workplace.

witnesses became more emotionally drained when they took on the perspective of the target and when they witnessed the mistreatment of a colleague directly, thus indicating that degree of identification with the target and proximity to the target are important moderators for the association between witnessing bullying and outcomes, possibly creating an even stronger cognitive dissonance.

Regarding moderating variables, a key finding from the meta-analytic part of the current review is that the magnitude of the association between witnessing bullying and mental health outcomes were dependent upon the witnesses' own exposure to workplace bullying. Specifically, we found that the effect size between witnessing bullying and mental health complaints were significantly smaller in studies that adjusted for the observers own exposure compared to studies that did not perform this kind of adjustment. This finding has several important implications. First, it indicates that previous research on outcomes among witnesses that have not performed this kind of adjustment most probably have overestimated the actual strength of associations, thereby providing reason to question the trustworthiness of the estimates. Second, the result adds to our understanding of the "victim by proxy" (also known as the "secondary victims") hypothesis which claims that witnessing mistreatment of others at the workplace is threatening for the observer as it may indicate how any employee may be treated (Sprigg et al., 2019). Having shown that witnessing bullying only seems to have a detrimental impact on witnesses who have been exposed to bullying themselves, our findings suggest that a victim by proxy effect requires that the witness has own previous knowledge about the impact of bullying and thereby identify with the target. Hence, it is likely to be personal recognition of mistreatment and a fear of re-experiencing bullying that explains the association between witnessing bullying and outcomes. Taken together, including the witnesses' own exposure to bullying either as a moderator or control variable seems to be a requirement for understanding potential outcomes.

As for other moderating variables that should be considered in research on outcomes among witnesses to bullying, Sprigg et al. (2019) showed that factors, such as coworker and supervisor support at the work group and organizational levels, are important with regard to determining the consequences following witnessed bullying. Furthermore, as mentioned previously, Nielsen, Rosander, et al. (2021) found that whether a witness intervene against the observed bullying or not may play an important role. Consequently, based on what we know about conditional factors, we have included the following moderating variables in our conceptual model (Fig. 2): The witnesses' own exposure to bullying, degree of identification with the target, proximity to the target, frequency of observations of mistreatment, intervention behavior from the witness, organizational interventions, and support from others in the organization. We believe that these factors may contribute to explaining the psychological and attitudinal outcomes in those witnessing bullying at the workplace. In addition, as suggested by Sprigg et al. (2019) and Totterdell et al. (2012), individual factors such as personality, values and empathy are also likely to play a moderating role, either as direct moderators of the association between witnessing and outcomes or as determinants of the impact of the situational moderators. Based on the evidence established in this review, these moderators should therefore be prime candidates for inclusion in future studies on witnesses to bullying. In addition, to move the field of research forward and increase our understanding of how witnessing bullying can have an impact on the observers, it is equally important to identify and test new and alternative moderating factors as well as testing more complex models with multiple moderators.

As shown in our conceptual model, *time* may also be a crucial factor for understanding the association between witnessing bullying and outcomes. While most previous research have relied on the assumption that a significant correlation between witnessing bullying and a given outcome indicates that being a witness to bullying leads to poorer health and well-being, the overuse of cross-sectional designs within this field of research limits any conclusions about such a causal effect. We found two

studies based on prospective data that provided evidence for a reverse relation in which mental health problems among the respondents increase the likelihood of witnessing bullying at a later time-point. Consequently, based on cross-sectional data we do not know whether an established association is explained by the fact that witnessing bullying causes health problems or whether it rather is explained by existing health problems leading to an increased risk of and sensitivity when witnessing bullying. To determine the direction of the relation between study variables, future research should rely on longitudinal designs that can shed light on the existence of reverse causality. Of course, the reverse relations, including their explanatory mechanisms, is an interesting venue for research on its own. Nielsen and Einarsen (2013) provided two theoretical explanations for a reverse relation: (a) Following the "gloomy perception mechanism" (de Lange et al., 2005), distressed employees may report less favorable work characteristics because they evaluate their work environment more negatively than do other employees; and (b) persons with mental health problems have a lower tolerance for exposure to bullying and, as a consequence, also have a lowered threshold for interpreting negative event at the workplace as bullying. Both these explanations should be empirically tested in upcoming research on witnesses to bullying.

Finally, it should be noted that all the included primary studies were limited to examine group difference between witnesses and non-witnesses regarding outcomes or linear relationships between witnessing bullying and outcomes. We can therefore not rule out the potential for non-linear associations. For instance, it could be that the initial observation of bullying may come as a substantial shock to the witness and therefore have a more extensive impact compared to later observations. That is, after first observing bullying additional observations will be experienced more as confirmation of the occurrence of bullying and may therefore have a lesser "shock-value". There may also be a subsequent dose-response relation where the impact of the witnessed bullying will only occur after a given number of observations. To determine such associations, upcoming research needs to take the frequency of observations into consideration and examine curvilinear and quadratic relationships.

4.2. Implications for practice

Our systematic review and meta-analytic findings have implications for organizational practitioners. The established relationships between witnessing workplace bullying and job-related outcomes such as job dissatisfaction and turnover intent show that bullying has detrimental consequences not only for exposed employees, but also for other employees at the organizations in which the bullying occurs. Hence, the findings support previous claims about bullying as a severe problem in the psychosocial working environment (Hauge et al., 2007; Van den Brande et al., 2016), and successful measures to prevent and reduce workplace bullying are therefore likely to also have positive effects for the organization as a whole.

As witnessing bullying seem to have limited impact on the health of witnesses that have not been exposed to bullying themselves, one important implication is that organizations may benefit from training employees to identify cases of bullying in their work group without imposing any risk of negative health consequences to such witnesses-to-be. That is, through increasing the employees' awareness of, knowledge about, and perceptions related to workplace bullying and harassment, as well as how to handle bullying as an active and constructive bystander, this may contribute to foster helping behavior and more efficient reporting of bullying (Einarsen, Nielsen, et al., 2020). This may have two main benefits. First, through helping colleagues exposed to bullying, the target may be saved from detrimental long-term health consequences themselves, and even if the attempt at intervening fails, the target will experience social support, something that in itself may buffer against the negative effects of bullying (Nielsen, Rosander, et al., 2021). Second, through encouraging witnesses to report the cases they have observed,

the organization will have an effective marker for timely intervention, something that can contribute to end the bullying at an early stage. Consequently, by promoting social support and constructive behaviors among bystanders, one may be able to reduce the overall occurrence of workplace bullying in the longer run (Einarsen, Nielsen, et al., 2020). However, the effectiveness of such an intervention will of course depend upon the existence of well-developed policies and procedures for handling cases of bullying (Duffy, 2009; Salin, 2008).

Lastly, active intervention when witnessing bullying of others is likely to make the problem more visible, which in itself can help to put the work environment on the agenda and encourage actions to improve it further (Rosander & Nielsen, 2023). However, to be effective, this means that any fear of intervening that may exist among employees needs to be dealt with in the organization, for instance through having available reporting procedures and ways of safeguarding employees that do report bullying (Rayner & Lewis, 2011).

4.3. Methodological limitations

The findings and conclusions of a systematic review and meta-analysis are always dependent on the quality of the studies included. As our quality assessment showed that the included studies varied regarding the methodological rigor, the reading of the review and its finding needs to be guided by an understanding of some important methodological limitations of the included studies and the review. First, despite a relatively general and broad research strategy that was considered to reduce the risk of selection and detection bias, we were only able to identify a small number of relevant studies. As the majority of the included studies examined some form of psychological distress as the main outcome, the knowledge base for other outcomes such as somatic complaints, sleep problems, commitment, and turnover intent is especially limited, and the established effect sizes are therefore associated with some uncertainty. Regarding the research strategy, it should be noted that the review included some studies that did not assess bullying explicitly, but rather described related topics such as incivility, harassment, and discrimination. Although such exposures may represent types of bullying behavior, we do not know whether the witnesses labelled what they saw as bullying. Still, as the findings from the studies on these related constructs were consistent with those that used the label “bullying”, this nevertheless indicates a certain overlap in scope.

Second, all studies were based on self-report data and a large share of the studies were based on cross-sectional data collected with non-random sampling procedures. These methodological shortcomings are likely to limit the external validity of the findings. Thus, there is a need for further prospective studies based on randomized and representative population samples with appropriate sample sizes, and preferably objective assessments, in order to improve the internal and external validity of research on the outcomes following witnessed bullying at the workplace. It should be noted that although prospective data often are considered as “better” than cross-sectional data since the former can provide indications of potential causal orders, there are also some important challenges when using prospective designs (Spector, 2019). For instance, it has been shown that effect sizes found in prospective studies are influenced by the utilized time-lag between the measurement points and that choosing the “correct” time-lag often is challenging (Dormann et al., 2015; Ford et al., 2014). Hence, by using a too short or too long time-lag there is a risk that an actual association between a pair of variables is not detected, something that can lead to type II errors. Cross-sectional data that can provide indications of a relationship may therefore be beneficial as a supplement to prospective data.

Third, this review was limited to published papers in peer reviewed scientific journals, thus excluding gray literature in the form of unpublished studies and non-peer reviewed literature (e.g., reports). Although it has been suggested that researchers should aim at including unpublished literature in meta-analyses and systematic reviews, the inclusion of data from unpublished studies can itself introduce bias (Higgins &

Green, 2011). First, unpublished studies that can be located are likely to reflect an unrepresentative sample of all unpublished studies (Higgins & Green, 2011). Secondly, unpublished studies may be of lower methodological quality than published studies. In a study of 60 meta-analyses that included published and unpublished studies, unpublished studies were less likely to conceal intervention allocation adequately and to blind outcome assessments (Egger et al., 2003). Hence, by only including peer-reviewed studies, the scientific quality of the included studies should be ensured at a minimum level at the same time being representative for the published literature on the outcomes following witnessed bullying.

5. Conclusion

Our review has shown that existing research and findings on outcomes among witnesses to workplace bullying are relatively fragmented and inconclusive. This may be due to the fact that witnessing bullying is a complex phenomenon and that the outcomes rely on a range of third variables and indirect relationships. That is, even if previous literature has found that witnesses to bullying at the workplace are likely to report poorer health and well-being than non-witnesses, we have shown that such a relationship is not necessarily as straight forward as one may expect. To understand the true consequences, we also need to consider reverse causation, the witness's own exposure to bullying, their proximity and identification with the target, as well as their helping behavior. Therefore, we encourage future research to (a) further integrate our proposed theoretical/conceptual framework into the study of witnesses, (b) investigate established and new moderators and mediators, and (c) study causal effects in the opposite directions than typically assumed.

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Data availability

Data will be made available on request.

Appendix A. Supplementary data

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